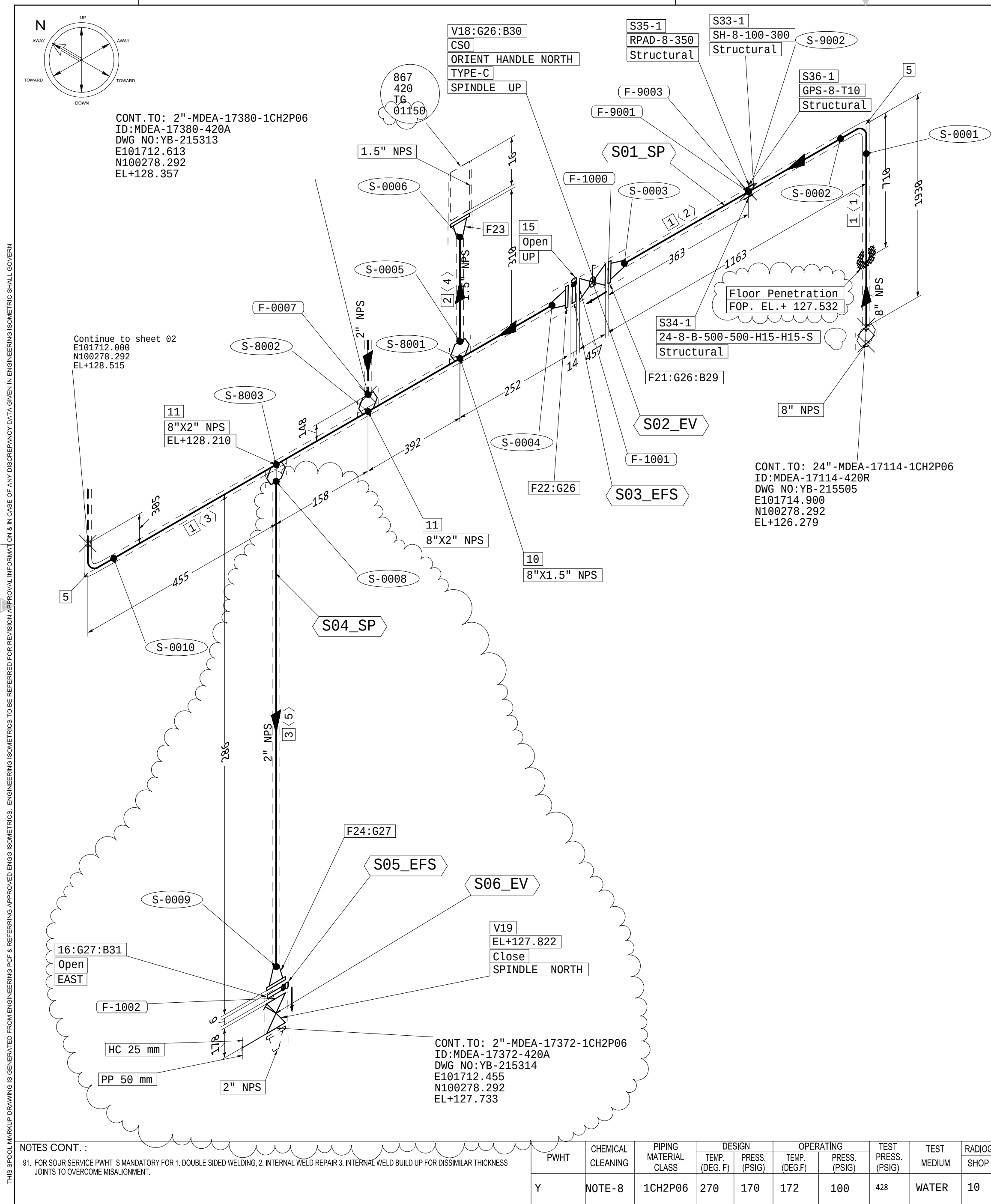


THIS SPLOO MARKUP DRAWING IS GENERATED FROM ENGINEERING P&ID & REFERRING APPROVED ENGG ISOMETRICS. ENGINEERING ISOMETRICS TO BE REFERRED FOR REVISION APPROVAL INFORMATION & IN CASE OF ANY DISCREPANCY DATA GIVEN IN ENGINEERING ISOMETRIC SHALL GOVERN



MATERIALS						STANDARD DETAIL REFERENCE DRAWINGS																								
PT No.	DESCRIPTION			NPD(IN)	ITEM CODE	QTY	DETAILS OF HEAVY WELDING BOSSSES FOR THREADED CONNECTIONS CONSTRUCTION/NOTES AND DETAILS FOR ISOMETRIC PIPING INSULATION DETAILS FOR HOT/THINWALL INSULATION DETAILS FOR COLD/THINWALL INSTALLATION OF JACK SCREWS FOR FLANGED JOINTS					AE-036175 AD-036627 AA-036911 AA-036913 AC-036930																		
	PIPES						NOTES 1. MATERIAL SHOWN WITH A BROKEN LINE AND SPOOLS LISTED IN THE SAUDI ARAMCO FABRICATION LIST ARE NOT TO BE FURNISHED BY THE VENDOR. 2. PIPING DETAILS SHALL BE CUT AT ALL POINTS MARKED F.W. (FIELD WELD). 3. EACH FABRICATED SECTION SHALL BE CLEARLY MARKED WITH ITS CODE NUMBER. 4. UNLESS OTHERWISE NOTED: (A) ALL DIMENSIONS ARE TO FACE OF FLANGES. (B) BOLT HOLES ON FLANGES STRAIGHT C. (C) WELDING ELBOWS ARE LONG RADIUS. 5. FABRICATION SHALL BE IN ACCORDANCE WITH SAUDI ARAMCO SPECIFICATION 01-SAMBS-10. 6. GASKET ALLOWANCE _____ 7. PIPING MATERIAL IS _____ <table><tr><th>PIPE SIZE</th><th>SCHEDULE</th><th>SPECIFICATION</th></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr><tr><td> </td><td> </td><td> </td></tr></table> 8. WELDING STRESS RELIEF - YES ☐ NO ☐ 100% X-RAY ☐ SPOT X-RAY 01-SAMBS-10 ☐ 9. TESTING SHOP HYDROSTATIC TEST - YES ☐ NO ☐ THE SHOP HYDROSTATIC TEST SHALL BE CARRIED OUT ON PIECES HAVING ALL ENDS FLANGED. HYDROSTATIC TEST PRESSURE: _____ 10. ALL PIPING MATERIAL IS IN ACCORDANCE WITH SAUDI ARAMCO ENGINEERING SPECIFICATION 01-SAMBS-10S CLASS _____ ALL EXCEPTIONS ARE NOTED ON THE ISOMETRIC. 11. INSULATION REQUIRED - YES ☐ NO ☐ INSULATION SHOWN ON DRAWING: _____					PIPE SIZE	SCHEDULE	SPECIFICATION																
PIPE SIZE	SCHEDULE	SPECIFICATION																												
1	PIPE SMLS 8" A106-B SOUR STD BE B36.10M	8	PPS2AERT08ZZABUAIA	3.232																										
2	PIPE SMLS 1 1/2" A106-B SOUR XS BE B36.10M	1.5	PPS2AERTAMZZABVAIA	0.105																										
3	PIPE SMLS 2" A106-B SOUR XS BE B36.10M	2	PPS2AERT02ZZABVAIA	0.075																										
5	ELB 90° SMLS LR 8" A234-WPB SOUR STD BE B16.9	8	PE92AZRT08ZZABUIAQ	2																										
10	WELT 8" X 1 1/2" A105N SOUR STD X XS BE SP-97	8X1.5	PWL2ADRT08AMAKRAI7	1																										
11	WELT 8" X 2" A105N SOUR STD X XS BE SP-97	8X2	PWL2ADRT0802AKRAI7	2																										
15	SPCT. BLD FF 8" A516 GR70 NORMALISED SOUR 150# 125 SPEC	8	PLF2RVRT08ZZAAAADZ	1																										
16	SPCT. BLD FF 2" A516 GR70 NORMALISED SOUR 150# 125 SPEC	2	PLF2RVRT02ZZAAAADZ	1																										
18	TAG NO V11-CB-RS-AJ-AB-AA-KQC BALL VAL RB API 60 8" A216-WCB SS316 TRIM, RTFE SEAT 2/3 PIECE CONST, TRUN MNT, FIRE SAFE RF 150# SOUR, LNG PATN + CSO/CSC, GEAR OPT	8	V11CBRS08AJABAAKQC	1																										
19	TAG NO V60-CB-AU-AH-AB-AA-SF6 GATE VAL API 600 2" A216-WCB TRIM 16 FLX WEDGE, BLT BNT, OS & Y RF 150# SOUR, HAND WHL OPT	2	V60CBAU02AHABAA5F6	1																										
FLANGES																														
21	FLG WNRF 8" A105N SOUR STD, 150# 125 B16.5	8	PFY2ADRT08ZZACPADH	1																										
22	FLG WNRF JACK SCR 8" A105N SOUR STD, 150# 125 B16.5	8	P6T2ADRT08ZZACPADH	1																										
23	FLG WNRF 1 1/2" A105N SOUR XS, 150# 125 B16.5	1.5	PFY2ADRTAMZZACQADH	1																										
24	FLG WNRF JACK SCR 2" A105N SOUR XS, 150# 125 B16.5	2	P6T2ADRT02ZZACQADH	1																										
GASKET																														
26	GSK SPR WND W/CS CENTERING RING + I - SS 8" 4.50 MM THICK SS316L+GRPH FILL SOUR 150# B16.20	8	P6UUB5RT08AIAAAZZC	3																										
27	GSK SPR WND W/CS CENTERING RING + I - SS 2" 4.50 MM THICK SS316L+GRPH FILL SOUR 150# B16.20	2	P6UUB5RT02AIAAAZZC	2																										
FASTSNERS																														
29	BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 120 mm Length	0.75	PH8ASCRTAG00120ZZF	8																										
30	BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 140 mm Length	0.75	PH8ASCRTAG00140ZZF	8																										
31	BOLT, STUD+2 NUTS A193 GR B7M/ A194 GR 2HM, CERAMIC FLP COAT SOUR B18.2.1/18.2.2 - 100 mm Length	0.625	PH8ASCRTAF00100ZZF	4																										
SUPPORTS																														
33	PIPE SUPPORT	8	SH-8-100-300	1																										
34	PIPE SUPPORT	8	24-8-B-500-500-H15-H15-S	1																										
35	PIPE SUPPORT	8	RPAD-8-350	1																										
36	PIPE SUPPORT	8	GPS-8-T10	1																										
PIPE CUT LENGTH SUMMARY																														
SPL No.	NPD	PC No.	LENGTH	ALLOWANCE	TOTAL LENGTH																									
S01_SP	8	<2>	0.757	0.000	0.757																									
S01_SP	8	<1>	1.625	0.000	1.625																									
S04_SP	2	<5>	0.075	0.000	0.075																									
S04_SP	8	<3>	0.850	0.000	0.850																									
S04_SP	1.5	<4>	0.105	0.000	0.105																									
NOTES :																														
1. ALL COORDINATES AND ELEVATIONS ARE IN 'METER' AND ALL DIMENSIONS ARE IN 'MM'. 2. PIPE QUANTITY IS IN METERS AND OTHER QUANTITY IN NUMBERS. 3. FOR STANDARD SUPPORT, REFER DOC. NO. "PP-868-ST-0001". 4. IN CASE OF ANY DISCREPANCY FOR LINE CONDITIONS, LINE DATA GIVEN IN LINE LIST SHALL GOVERN. 5. FOR SMALL BORE LINES, DIMENSION & ROUTING SHALL BE FIELD ADJUSTED. 6. THE SPECIFIED THEORETICAL LOCATION OF EQUIPMENT NOZZLES, SUPPORTING STRUCTURE SHALL BE VERIFIED BEFORE INSTALLING THE PIPING & INTOLERABLE DISCREPANCIES SHALL BE DISCUSSED WITH DESIGN SITE ENGINEER. 7. ALL THREADED CONNECTION SHALL BE SEAL WELDED, EXCEPT DRAIN AND VENT PLUG. 8. REFER LINE LIST. 9. ALL PRESSURE WELDS, OTHER THAN BUTT WELDS, INCLUDING BRANCH CONNECTIONS SUCH AS BRANCH-OLETS, AND BOSS) SHALL BE 100% PTMT EXAMINED. ALL ATTACHMENT WELDS ON PRESSURIZED COMPONENTS THAT ARE NOT HYDROTESTED AFTER MAKING THE ATTACHMENT WELD SHALL BE 100% PTMT EXAMINED.																														
GRAPHY %			INSULATION	THK. (MM)	TYPE																									
FIELD	10	25	HC	NA	APCS-2A																									
STRESS SYS NO.			LINE ID																											
867-GPT4-154			MDEA-17357-420A																											
BUTT WELD FIELD WELD INSULATED PIPE TRACED PIPE PIPE SUPPORT PLATFORM PENETRATION SOCKET WELD SCREWED JOINT																														
DESIGN CERTIFICATION (DESIGN AGENCY) REVIEW FOR KEY DRAWINGS (SAUDI ARAMCO) OTHER																														
ENGL DATE		CERTIFIED BY DATE		CONS/ AGENY DATE																										
OPERATIONS DATE		BY DATE																												
REV NO.		DATE		REVISION DESCRIPTION																										
JOB ORDER/EWO NO.																														
10-14815-0003																														
SAUDI ARABIAN OIL COMPANY																														
ISOM 8" -MDEA-17357-1CH2P06																														
MDEA SERVICE																														
JFGP GAS PROCESSING TRAIN-4																														
JAFURAH SAUDI ARABIA																														
PLANT NO.		INDEX		DRAWING NUMBER																										
867		L		YB-215304																										
SHT. NO.		REV. NO.																												
010f05		00A																												
SITE REV. NO.		00A		RESTRICTED SMART																										
CADD-05																														

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